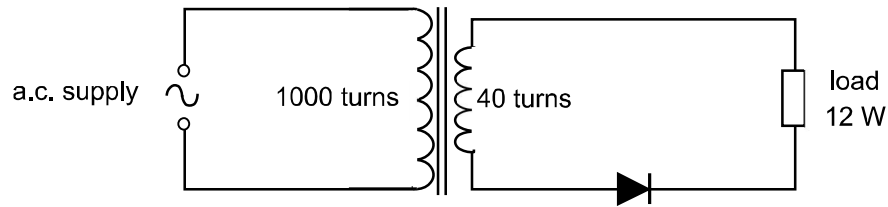


- 28** The primary coil of an ideal transformer has 1000 turns and is connected to a sinusoidal a.c. supply. The secondary coil has 40 turns and is connected to an ideal diode and a load.



If the root-mean-square (r.m.s) voltage across the load is 6.5 V and the mean power dissipated in the load is 12 W, what is the r.m.s current in the primary coil?

A	0.026 A	B	0.074 A	C	0.10 A	D	0.15 A
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Answer: C

On the secondary side, at the load, $I_{L\text{ rms}} = \frac{\langle P \rangle}{V_{L\text{ rms}}} = \frac{12}{6.5} = 1.8462\text{ A}$

Since the current at the load is Half-wave rectified, it can be shown that

$$I_{L\text{ rms}} = \sqrt{\frac{I_{L\text{ peak}}^2}{4}} = \frac{I_{L\text{ peak}}}{2} \Rightarrow I_{L\text{ peak}} = 2 \times 1.8462 = 3.6923\text{ A}$$

On the secondary side before the diode, the current is Full-wave sinusoidal A.C.,

$$I_{S\text{ rms}} = \frac{I_{S\text{ peak}}}{\sqrt{2}} = \frac{3.6923}{\sqrt{2}} \Rightarrow 2.6109\text{ A}$$

Therefore using

$$\frac{I_P}{I_S} = \frac{N_S}{N_P}$$

$$I_P = \frac{40}{1000} \times 2.6108 = 0.10443 = 0.10\text{ A}$$

